

Raghav Senthil Kumar

623-332-2168 | raghavs9@illinois.edu | linkedin.com/in/raghav | github.com/raghav

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Designed and implemented a text-retrieval algorithm that validates source URLs and filters outliers with **IQR**
- Reduced search index bloat by flagging **30%** of evidence documents as identical with a de-duplication algorithm
- Scaled search to **600K+** debate-evidence documents on *Elasticsearch*, returning results in **under 1 second**
- Collaborated with **10 engineers** in an *Agile* team on an *AWS*-hosted SaaS platform drawing **10K+** site visitors

Product Development Intern | Jetson

January 2025 – February 2025

- Documented user-engagement findings and recommendations clearly in a **5-page report** directly for the CEO
- Evaluated **3 experimental features** through **7 iterations of UI/UX testing** in an app with **100K+** users
- Modified the onboarding flow in *Figma* to cut it from **12 to 5 pages**, then validated the change with **A/B testing**

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed **200+ TikTok posts** as **3 years** of time-series data with *Python (pandas)*, *Power BI* and *Microsoft Excel*
- Identified a clear pattern: “science fun facts” posts drew **69.3%** of views from just **7%** of all posted content
- Recommended a “fun facts” strategy the Executive Director adopted, improving engagement **24%** (8K+ views)

PROJECTS

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Applied **4 Random Forest models** to detect anomalies in live water, waste and energy data from all **3 tools**
- Turned anomaly detections into prescriptive, actionable steps via a *Gemma* Voice LLM on a *Next.js* dashboard
- Built a *Python FastAPI* back end that ingests water, waste and energy data from all **3 tools** for anomaly detection
- Developed **3 edge AI tools** that collect data on a school’s energy, water and food use for anomaly detection

EcoBin | Research Paper | [Source Code](#)

March 2026 – May 2026

- Trained a deep learning model (*EfficientNetV2-S*) in *PyTorch* on **15K+** images to reach **96.13% accuracy**
- Published the research as an *arXiv* paper with **2 datasets** on *Kaggle*. **2nd at Arizona State Science Fair**
- Built a synthetic data pipeline (*U2-Net* segmentation, alpha-compositing) producing **9K** contamination images
- Engineered a smart bin prototype on an *Arduino Uno* that sorts each waste item with **under 2 seconds** of latency

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Built a predictive *quantile regression* model that estimates truck prices with **87% accuracy** on **30 samples**
- Ran root-cause analysis on each wrong prediction, revising the schema until **10 consecutive images** were correct
- Audited data quality across **15,000** truck-listing images by de-duplicating listings and filtering non-USD prices
- Structured photo data with a *JSON schema* that guides a *Claude Sonnet 4.6* agent to extract truck details
- Won **2nd Place at UIUC Founder’s Hackathon** leading team of **4** on pricing algorithm for Kamion (YC S22)

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Built a dual-API ETL pipeline that ingests papers from *OpenAlex* and backfills metadata from *Semantic Scholar*
- Modeled paper data in *PostgreSQL* on *Supabase* and queried it with *SQL* by field, year and total citation count
- Added a **RAG** agent (*Claude Sonnet 4.5*) that answers from session-scoped vector documents via *Supernormality*
- Developed a web UI on *React Flow* that graphs each paper’s outbound citations across **26 research fields**

SKILLS

Programming Languages: Python, SQL, Java, C++, JavaScript, HTML/CSS

AI/ML: scikit-learn, TensorFlow, PyTorch, Keras, OpenCV, Grad-CAM, Hugging Face, Claude API, Gemma 4

Data & Analytics: pandas, NumPy, Power BI, Microsoft Excel, A/B Testing, Hootsuite

DevTools: Git, GitHub, Kaggle, Docker, CI/CD, VS Code, Claude Code, Codex, Figma, Bun, Clang, Catch2

Concepts/Practices: ETL, LLMs, GenAI, Event Classification, Anomaly Detection, Statistical Analysis, Clustering